how the new model formed a series of rather original evaluations of the problem that was presented to it in a prompt, broke it down into simpler segments and then produced a response that it thought would be the most accurate for its user.

This ties into our next two points, which are about the increased accuracy of model o1 to open-ended prompts and reasoning queries.

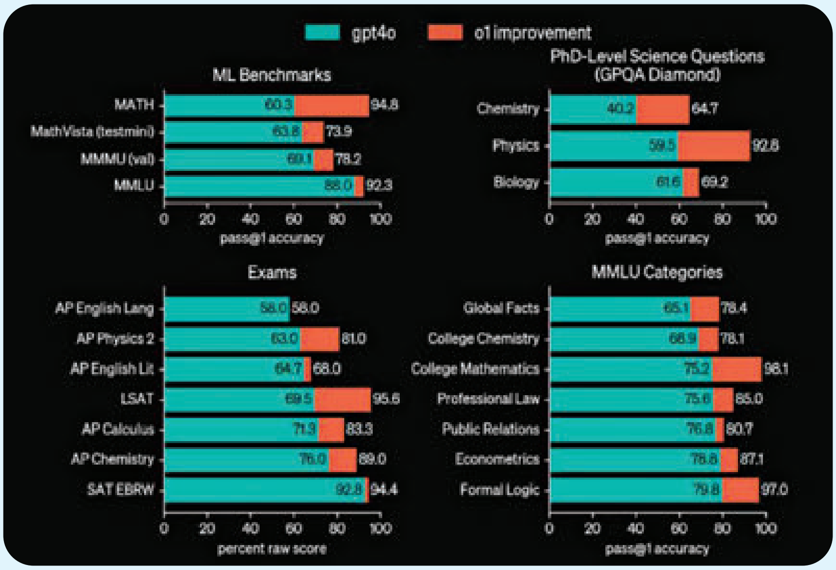

Image Source: OpenAI

GPT-4o vs OpenAI o1 model

Better response to open-ended prompts – That's what the humans say

During the development of the new model, OpenAI tested the efficacy of model o1 when working with open-ended prompts. Not all of us are prompt engineers. If you have worked with generative AI models that are available today, you would know that, more often than not, open-ended prompts are not handed well by the existing models. After generating a response with an open-ended prompt, users are forced to refine the response by feeding in follow-up prompts.

With model o1, OpenAI claims that the responses generated by the model in response to open-ended prompts are preferred by humans over the ones generated by GPT-4o. However, we hit a roadblock when it comes to natural language processing and related applications, as OpenAI, in their documentation about the model, mentioned – "o1-preview is not preferred on some natural language tasks, suggesting that it is not well-suited for all use cases."

OpenAI's Strawberry has increased accuracy in reasoning queries

After thoroughly testing model o1, OpenAI concluded that the new model is consistent with generating more accurate results with respect to reasoning queries. The model apparently even outperformed human experts in certain benchmarks that were run while evaluating the model. These benchmarks spanned across multiple fields that have been pioneered by humans, including the likes of chemistry, physics and biology.

Not to mention, model o1 is better than GPT-4o in reasoning-heavy tasks. To back this claim, OpenAI has mentioned that the GPT-4o was able to solve an average of 12 per cent (1.8/15) of problems in the 2024 American Invitational Mathematics Examination (AIME). When working with a single sample per problem in the same set of questions, model o1 maintained 74 per cent accuracy, answering 11.1 correctly out of the 15 it was presented with. When the sample set was expanded, the model achieved an accuracy of up to 93 per cent.

Strawberry is a better coder than you are

In the benchmarking done by OpenAI, Strawberry achieved an ELO of 1807 compared to the 808 achieved by 4o. Open AI went a step ahead and trained a model to participate in a simulation run of the 2024 International Olympiad in Informatics (IOI), where it was pitted against human contestants and had ten hours to solve six algorithmic problems with the submissions for each problem being limited to 50. When the tests were run, model o1 was able to rank in the 49th percentile with a score of 213 points to its name. The documentation about the model dives further into the strategy it employed during the run: "For each problem, our system sampled many candidate submissions and submitted 50 of them based on a test-time selection strategy. Submissions were selected based on performance on the IOI public test cases, model-generated test cases, and a learned scoring function. If we had instead submitted at random, we would have only scored 156 points on average, suggesting that this strategy was worth nearly 60 points under competition constraints."

What's next in the future of OpenAI Model o1?

The increased capabilities of model o1 over its predecessor, as OpenAI tries to perfect it, opens doors to a future that would seem rather scary on the surface; however, when you dive deeper, it has a sea of possibilities. If models like this are able to perform with increased efficiency and consistency, a lot of data-driven evaluation and problem-solving tasks across fields of research and development could be off-loaded to an AI model, as humans divert their attention towards tasks that would involve critical thinking by humans. d